

Side-By-Side Might Win: Occlusion Negatively Affects The Performance Of Augmented Reality Task Instructions

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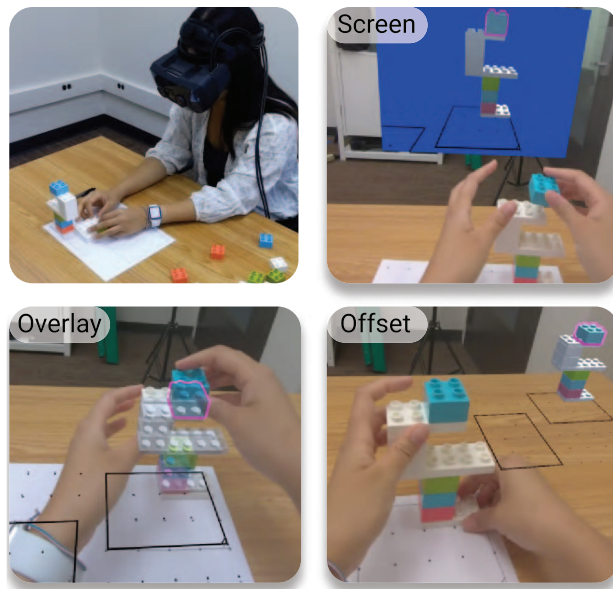


Figure 1: Images of one of the Lego blocks and drawing tasks with AR guidance: Each of the three different AR visualization modes for the same step in the same task is shown.

ABSTRACT

We compare different representations for Augmented Reality instructions for a combined pick & place and drawing task. Participants completed the task with instructions presented as 3D AR overlays, as 3D AR offset (side-by-side), or as 2D panels. Results indicate that participants preferred the 3D offset instructions, especially compared to 3D overlays. Our results stand in contrast to prior work where 3D overlays were preferred. Our work points at the need for the community to better define benchmarks and standardized tests to create guidelines for when to use what type of AR representation.

1 INTRODUCTION

Augmented Reality (AR) is frequently proposed to present users with instructions and tutorials for a wide range of tasks such as furniture assembly, pick, and place, task instructions, or maintenance [5, 4]. With the technology, a wide range of representations for the instructions have been proposed, including 2D, 3D, abstract vs concrete, smartphone-based, in-site as overlay, or side-by-side. We refer the reader to work by Jansche et al. [4] for an overview of representations. In general, differences in tasks, hardware, and im-

plementation influence which representation is considered optimal in terms of task performance and subjective ratings [6]. We contribute to this literature by providing further, and partly conflicting results, as to what representation of task instructions are optimal. The goal of this work is to foster discussion in the community on the influence of various factors and start a conversation about why there seems to be little agreement in the literature.

We explore the usage of AR instructions for medium-complex tasks, specifically users' preference on the representation of instructions in the AR task for sequential tasks in a controlled experiment ($n = 12$). We compare a traditional 2D floating-window baseline with 3D instructions that are directly overlaid on top of the physical objects, and with 3D instructions that are shown offset from the main building area. Participants exhibited a preference for the side-by-side AR instructions. Instructions directly overlaid onto the physical targets led to occlusion and were less well received by participants.

We like to note that this directly contradicts Blattgerste et al. [1], who found that AR overlay (in-situ) is preferred. We believe that this is due to slight variations in the task (combined Lego + paper-based task), variations in hardware (HoloLens vs Varjo XR-3), or other factors that are not well controlled for. This points to a much larger design space and the fact that without standard hardware and benchmark tasks, there cannot yet be any conclusive design guidelines on optimal AR representations.

2 STUDY: AR REPRESENTATION

Our goal is to find an optimal representation for AR task instructions. As a task, participants simultaneously built a Lego model and completed a drawing task, shown in Figure 1.

2.1 Study design

We employed a within-subject design with the independent variable *representation* with three levels: screen, AR offset, and AR overlay. The instructions were controlled by an expert in a wizard-of-oz manner. The order of representations was counterbalanced using a Latin Square.

2.1.1 Tasks

We created three different tasks to build a model with LEGO DUPLO blocks on a paper grid. In addition to constructing the blocks, participants were asked to draw specific patterns and connections on the paper where the blocks were placed. For each task, different rules of how to and what to draw depending on the positions of the Lego bricks were defined (e.g., connect red blocks with line). The purpose of those rules for drawing was to increase the complexity of the task, mimicking real-world tasks such as maintenance or multi-part assembly.

2.1.2 AR instructions

Participants were presented with AR instructions showing where to place blocks. Instructions were controlled by an expert in a wizard-of-oz manner. We varied the representation of the AR instructions (shown in Figure 1). The *screen* representation is a virtual screen with a fixed world position, showing one static image per instruction. The *AR offset* representation shows a 3D model of the target at a

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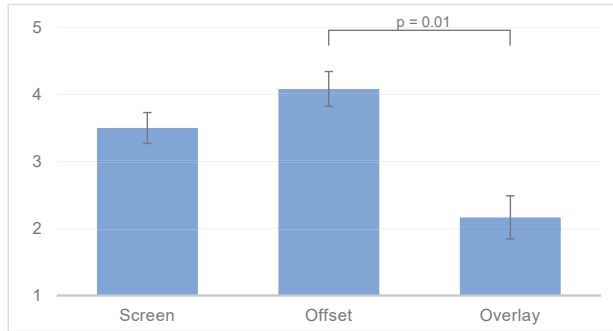


Figure 2: Participants' preference of *AR visualization* from 1 (strongly disagree) to 5 (strongly agree) for the question "I prefer using this guiding mode.". Error bars indicate standard error.

fixed world position next to the physical building location. The *AR overlay* representation uses the same model as the *AR offset* but displays it as an overlay with transparency.

2.2 Participants & apparatus

We recruited twelve participants from a local university, aged $M = 21.2$ years ($SD = 2.5$). Participants reported an average experience of $M = 2.4$ ($SD = 1.0$) with AR/VR headsets, with two having had no experience at all. The experiment was conducted in a quiet experimental space. We used a Varjo XR-3 headset to present AR instructions. The experimental software ran on a gaming computer (Alienware R12). Spatial calibration is performed manually to align the physical task and AR content.

2.3 Procedure & data collection

Participants first completed a demographic questionnaire and were then introduced to the instructions. Participants subsequently completed the three physical tasks. For each of the tasks, one of the three different representations was used, counterbalanced using a reduced Latin Square. For each task, the experimenter first explained what to draw depending on what the Lego build looked like. Next, the guidance with the AR headset started, letting participants first build bricks and then draw. Participants and the expert were recorded with webcams during the study. After each condition, they were asked questions from the NASA TLX [3] (stress) and SUS [2] questionnaires (use system frequently; easy to use) on scales from 1 (strongly disagree/very low) to 5 (strongly agree/very high). After the three physical tasks, participants were asked what representation they preferred. We did not analyze task completion time, since participants were instructed to perform the task at a reasonable pace (i. e., not focused on speed), and the expert instructions led to very few errors.

2.4 Results

2.4.1 AR Guidance & Representation

We performed a series of Friedman tests on the questionnaire data. Results indicate a statistically significant difference in participants' preferences of visualization modes ($\chi^2(3) = 10.711, p = 0.005$). Results are shown in Figure 2. Conover's post hoc comparisons with Bonferroni adjustment indicated statistically significant differences between the *AR offset* and the *AR overlay* visualization ($p = 0.010$). For the other questions, results revealed main effects for stress ($\chi^2(2) = 12.286, p = 0.002$), frequency of use ($\chi^2(2) = 14.368, p < 0.001$), and ease of use ($\chi^2(2) = 10.821, p = 0.004$), shown in Figure 3.

Nine participants referred to the *AR offset* visualization. Comments revealed that this was because it did not block the view of the build, and provided 3D information observable from every angle. Participants also appreciated the 2D *screen* visualization, because it

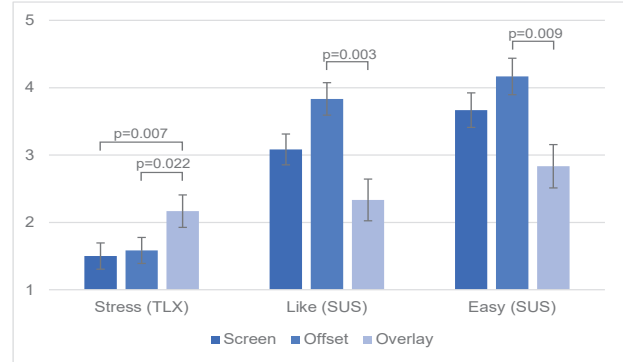


Figure 3: Results from Post-condition questionnaires from 1 (strongly disagree/very low) to 5 (strongly agree/very high). For Stress, lower is better; For "like" and "easy", higher is better.

was simple, familiar, and did not occlude the build area. The difference in preference between *AR offset* and *Screen* was not significant, indicating that both are viable alternatives. Participants found the *AR overlay* more irritating and harder to use, largely because the transparency of the 3D model made the virtual objects hard to see, especially their colors. It also occluded the real world, making placing blocks harder.

3 DISCUSSION, FUTURE WORK & CONCLUSION

The task in our experiment was a slight modification of the usual pick & place tasks from the literature [1, 6, 7]. In contrast to work by Blattgerste et al. [1], participants strongly preferred *AR offset* (side-by-side) in comparison to *AR overlay* (in-situ). We believe that both methods and experiments were sound, but that extraneous variables influenced those results. Blattgerste et al., similar to us, found results that were not in line with related work, where side-by-side typically exhibited higher performance or better subjective ratings. They argue that the difference might be because of improved tracking. We use tracking with similar performance, and a high-quality video pass-through headset. We speculate that the main contributing factor was the slightly more complex task. Nevertheless, we hope that our work inspires future work that establishes a benchmark setup to evaluate what representation is optimal. We have no doubt that the results will vary across modalities and factors, and hope that future work can establish better guidelines that reflect those results.

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